

The Functional Inheritance Model: Sequence-Level Grammar as Generative Constraint in Protein Organization

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Abstract

Analysis of 2,000 PDB structures, with filtered subsets of obligate homodimers (80,029 interface pairs) and SIFTS-confirmed enzyme interfaces (138,999 interface pairs), reveals a directionally consistent pattern of organizational signal across hierarchical levels. The diagnostic pair A→N shows Z-scores of −87.13 at the sequence level (Pfam functional domains, Anchor-Single sequential window method), −8.80 at enzyme non-interface surfaces, and −2.91 at enzyme protein-protein interfaces (direct spatial contact method). These measurements are derived from methodologically distinct analyses and are not directly comparable in absolute scale. After simultaneous ordinary least-squares regression against six biophysical and substitutional controls, 97.7% of the A→N signal at the sequence level remains, demonstrating near-complete orthogonality to known amino acid chemistry. The directional consistency across independent measurement approaches — signal strongest where organizational rules are established, attenuating where they are executed — is consistent with the Functional Inheritance Model proposed here, in which sequence-level grammar pre-positions viable organizational solutions that are inherited and utilized at the structural level. This model generates testable predictions including rapid adaptive responses on ecological timescales through selection among pre-existing grammatical possibilities.

1. Introduction

Standard accounts of molecular evolution assume that functional constraint intensifies at sites of direct biological function: protein-protein interfaces and enzyme active sites. Under this view, sequence-level patterns are presumed to be weak, diffuse, or merely neutral background. The present analysis tests this assumption directly across three organizational levels using independent measurement approaches.

1.1 The Conventional Prediction

If functional constraint concentrates at the site of direct biological function, the expected signal strength ordering would be:

Level	Predicted Signal Strength
Interface	Strongest
Surface	Moderate
Sequence	Weakest

The data presented here are inconsistent with this prediction. The organizational signal is strongest at the sequence level and attenuates at the structural levels. This pattern is interpreted as consistent with an alternative model described in Section 5.

2. Methods Summary

2.1 Datasets

Dataset	N	Method
Full PDB set	2,000 structures (Interactome3D 2024_12)	Sequential window (Anchor-Single)
Obligate homodimers	80,029 interface pairs; 159,274 non-interface	Sequential window (Anchor-Single)
Enzyme interfaces (SIFTS)	138,999 interface pairs; 802,673 non-interface	Sequential window (Anchor-Single)
Sequence reference	61,320,087 pairs (Pfam vs DisProt)	Sequential window (Anchor-Single)
Spatial contacts	842,498 inter-chain; 4,243,476 intra-chain	Direct atomic contact (5.0Å cutoff)

Enzyme structures were identified using SIFTS PDB-to-EC mapping (EBI, 2026 release), yielding 135,245 confirmed enzyme PDB IDs cross-referenced against the Interactome3D representative dataset. 6,292 unique enzyme chain pairs were used.

2.2 Diagnostic Pair

The A→N pair was selected as a diagnostic probe based on its extreme Z-score at the sequence level (−87.13), representing a strongly avoided substitution in functional domains relative to disordered regions.

2.3 Biophysical Controls

Regression controls: Kyte-Doolittle hydrophobicity difference, formal charge difference, molecular weight difference, Chou-Fasman α -helix propensity, Chou-Fasman β -sheet propensity, and BLOSUM62 substitution score. All features standardized before regression. Method: ordinary least squares.

Methodological note: The sequence-level and structural-level analyses use different methods, different datasets, and different null models. The sequence-level Z-scores derive from 61 million Anchor-Single pairs comparing Pfam functional domains to DisProt disordered regions. The structural Z-scores derive from direct spatial contact analysis of PDB structures with intra-chain surface contacts as the comparison set. These measurements are not directly comparable in absolute scale. The directional pattern across levels is the finding of interest.

3. Results

3.1 Directional Pattern Across Levels

Table 1 presents A→N Z-scores across organizational levels. These measurements come from methodologically distinct analyses (see Section 2.3 note). The directional pattern — stronger avoidance at the sequence level, attenuating at structural levels — is consistent across all subsets tested.

Level	Context	A→N Z-score	Mean Z	Role (proposed)
I. Sequence	Pfam Functional Domains	−87.13	High	Grammar established
II. Surface	Enzyme Non-Interface	−8.80	4.165	Rules expressed
III. Interface	Homodimer Interfaces	−2.30	1.648	Rules utilized
III. Interface	Enzyme Interfaces (SIFTS)	−2.91	1.872	Rules utilized

Table 1. A→N Z-scores across organizational levels from methodologically independent analyses. Absolute magnitudes are not directly comparable across levels due to differences in method, dataset, and null model. The directional pattern (stronger avoidance at sequence level) is the primary finding. Enzyme results use SIFTS-confirmed EC classification.

3.2 Signal Strength Comparison

Context	Interface Mean Z	Surface Mean Z	Ratio (I/S)
Full PDB set (2,000)	1.332	2.671	0.499
Obligate homodimers	1.648	2.020	0.816
Enzyme interfaces (SIFTS)	1.872	4.165	0.449

Table 2. Signal strength across subsets. Enzyme results reflect SIFTS-confirmed classification (6,292 unique chain pairs). In every subset the interface-to-surface ratio is below 1.0.

3.3 Orthogonality to Known Biophysics

After simultaneous regression against six physicochemical and substitutional controls, 97.7% of the sequence-level A→N signal remains ($R^2 = 0.0995$). The residual Z-score of −87.33 after controls essentially equals the raw value of −87.13, confirming near-complete orthogonality to all tested biophysical properties. This result applies specifically to the sequence-level analysis. At the structural level, biophysical controls explain a larger fraction of the signal (interface $R^2 = 0.513$), reflecting that direct contact chemistry contributes substantially to structural pair preferences.

3.4 Bonferroni Significance Patterns

Subset	Sig. at Interface	Sig. at Surface
Full PDB set	5/380	89/380
Obligate homodimers	32/380	49/380
Enzyme interfaces (SIFTS)	44/380	149/380

Table 3. Bonferroni-corrected significant pairs ($\alpha = 0.05/380 = 1.32 \times 10^{-4}$). Surfaces yield more significant pairs than interfaces in all subsets.

4. The Efficiency Gradient

Taken together, the measurements across three levels are consistent with a pattern in which organizational signal is strongest at the sequence level and attenuates progressively at structural levels. This pattern is observed across all subsets tested and survives the SIFTS-corrected enzyme classification.

This pattern is consistent with — though not proven by — the interpretation that the heaviest organizational work occurs upstream at the sequence and domain level, with downstream stages inheriting and utilizing that structure with progressively more selective but less extensive constraint. The preference concentration analysis supports this interpretation: the top-10 preferred pairs carry 16.5% of preference signal at the sequence level, rising to 29.1% at intra-chain surfaces and 54.5% at inter-chain interfaces. The Gini coefficient increases monotonically from 0.46 (sequence) to 0.64 (interface), indicating that organizational signal concentrates in a smaller set of chemically coherent pair types at the interface level.

Interpretive caution: The preference concentration at interfaces is dominated by charge-complementarity pairs ($E \leftrightarrow R$, $D \leftrightarrow R$, $E \leftrightarrow K$, $D \leftrightarrow K$), which are well-characterized biophysical interactions. The biophysical regression explains 51.3% of interface signal, substantially more than the 10% explained at the sequence level. The interface concentration finding is real but reflects known chemistry becoming dominant as well as any organizational inheritance effect.

5. The Functional Inheritance Model

The Functional Inheritance Model is proposed as an interpretation consistent with the data rather than a conclusion proven by it. It makes specific testable predictions that distinguish it from the standard gradualist account.

5.1 Core Propositions

Proposition 1 (Non-redundant Hierarchy). Organizational constraint is established at the sequence level and inherited downstream. Interfaces utilize pre-positioned materials rather than regenerating the grammar independently.

Proposition 2 (Pre-positioned Feedstock). The 97.7% residual signal after physicochemical controls demonstrates that viable organizational solutions exist within the sequence-level grammar. Whether this constitutes a pre-positioned feedstock for adaptation is an empirical question the model generates, not a conclusion the current data establishes.

Proposition 3 (Efficiency Through Inheritance). The directional pattern — stronger signal upstream, attenuating downstream — is consistent with a non-redundant hierarchical system. The same pattern could arise from other mechanisms including differential selective pressure across organizational levels.

5.2 Comparison with Standard Model

Standard Gradualist Account	Functional Inheritance Model
Strongest constraint at interfaces	Strongest constraint at sequence level
Interfaces generate novel grammar	Interfaces inherit pre-existing grammar
Adaptation requires new mutations	Adaptation may reveal latent grammatical possibilities
Incremental, memoryless variation	Variation on pre-structured substrate

Table 4. Standard model vs Functional Inheritance Model.

5.3 A Testable Prediction

The model generates a testable prediction: if sequence-level grammar pre-positions viable organizational solutions, adaptive responses to environmental change should be observable on ecological timescales without requiring new mutations. Darwin's finches — where beak morphology changes measurably within one to two generations following environmental pressure — are consistent with this prediction. Whether this reflects selection among pre-existing grammatical possibilities as the model proposes, or other mechanisms, remains to be established through direct empirical investigation connecting sequence-level organizational constraints to morphological outcomes. This connection is a prediction the model generates, not a demonstrated result.

6. Discussion

6.1 What the Data Establish

The data establish three findings with confidence. First, sequence-level amino acid pair preferences contain structured signal that is 97.7% orthogonal to six standard physicochemical controls at the diagnostic A→N pair. Second, this signal shows a directionally consistent pattern of attenuation from sequence to surface to interface across all subsets tested. Third, the preference architecture is coherent across structural levels (Pearson $r = 0.923$ between interface and surface Z-scores), indicating that the same pairs are generally preferred and avoided at both levels.

6.2 What the Data Do Not Establish

The data do not establish that the gradient reflects causal inheritance of organizational grammar from sequence to structure. The measurements come from methodologically distinct analyses and the directional pattern, while consistent, could reflect differential selective pressure, sampling differences, or other factors not captured by the current framework. The Functional Inheritance Model is a coherent interpretation of the pattern; it is not the only interpretation.

6.3 Predictions for Future Investigation

The model generates several testable predictions: (1) the 97.7% residual should replicate across independent protein families and fold types; (2) interfaces that are evolutionarily young should show weaker signal than ancient interfaces; (3) the directional pattern should hold in datasets independent of Pfam and Interactome3D; (4) rapid adaptive responses in other systems should be explicable through selection on pre-existing grammatical possibilities. These predictions distinguish the model from the standard account and are testable with available data.

7. Conclusion

The Anchor-Single pair analyses reveal a directionally consistent pattern across sequence and structural levels: organizational signal is strongest at the sequence level and attenuates progressively at structural levels. The sequence-level signal is 97.7% orthogonal to six biophysical controls at the diagnostic A→N pair. These findings are consistent with the Functional Inheritance Model, in which sequence-level grammar pre-positions viable organizational solutions that are inherited and utilized at the structural level.

The data do not identify the ultimate source of the sequence-level grammar, nor do they prove that the gradient reflects causal inheritance rather than other mechanisms. They demonstrate that functional protein sequences operate under a structured, non-random framework that is strongest at the sequence level and that current physicochemical models are substantially insufficient to account for.

The methodological contributions of this work — the Anchor-Single pair framework, the biophysical hardening procedure, and the cross-level directional analysis — provide tools for further investigation into the origin and maintenance of biological organizational grammar.

Data Availability

All data are derived from Interactome3D (2024_12), the RCSB Protein Data Bank, Pfam (functional sequences), DisProt (disordered sequences), and SIFTS EC mapping (EBI). Full datasets and analysis code are available from the Natural Organization Institute upon reasonable request.

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